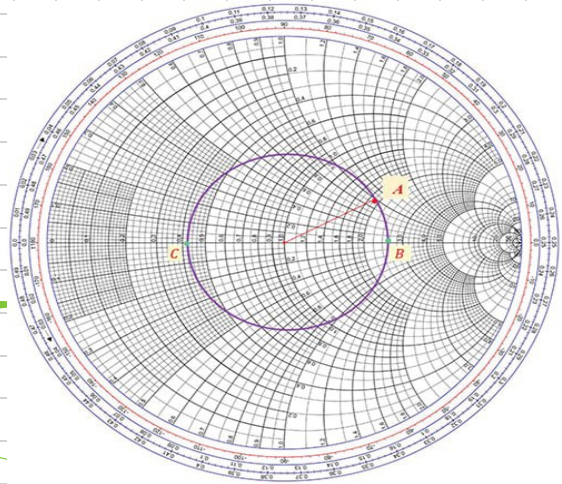
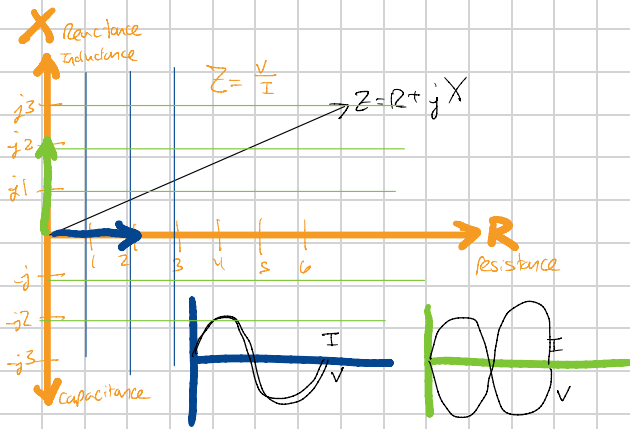


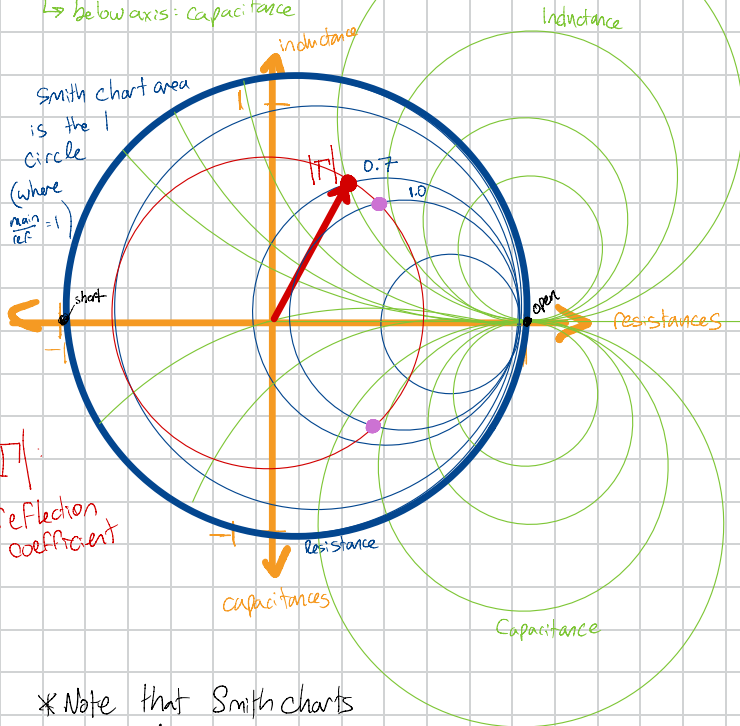
SMITH CHARTS



Pure Resistance
Pure Reactance

↗ above axis: inductance

↘ below axis: capacitance



$|\Gamma|$:
reflection coefficient

Impedance varies thru R and X on a transmission line, but magnitude is stable; Following the **reactance circle** will show all values across a wavelength.

By adding C or I , we can walk our circle to the **center of the chart** where $R=1$.

Ex: 50Ω transmission line, 108.5 MHz
in case of our $|\Gamma|$ in the diagram, we can add $\pm$ inductance by lengthening the line - if 12° rotation, calculate the length based on λ :

$$3.1 \text{ mm (or 2nd spot, } 102^\circ = 22 \text{ mm)}$$

$$1.8 = \frac{\omega L}{Z_0} = \frac{2\pi f L}{50\Omega} \quad L = 13.2 \text{ nH}$$

22mm length in this case.

*Note that Smith charts vary by frequency, and there are also admittance-based Smith charts.